Building Science

Lecture 8: Natural Lighting & Daylight

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Lecture outline

- Physics of Light
- Benefits of Daylight
- Daylight factor

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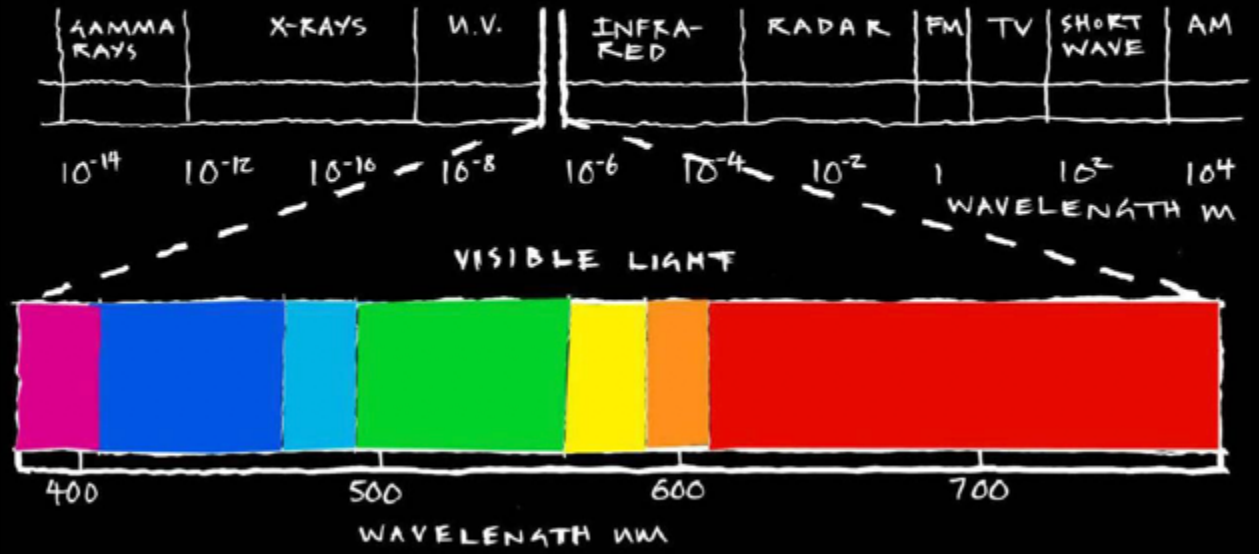
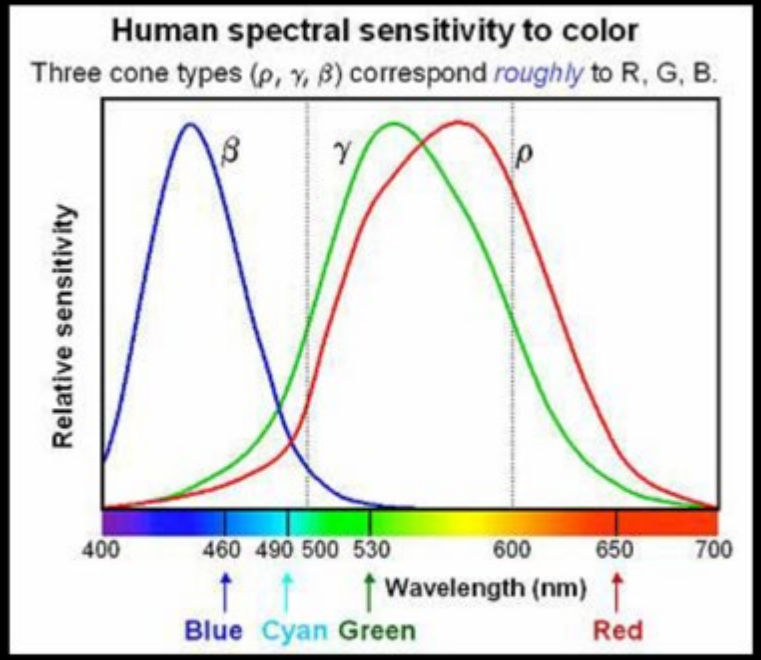
Physics of light

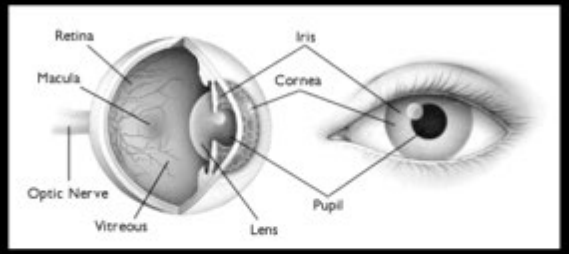
- ✓A narrow wavelength-band of electromagnetic radiation **(from about 380nm** to **780nm**) is perceived by human eyes



The electromagnetic spectrum

Response of the eye to different wavelengths of light





Three purposes of daylight design

1. Occupant satisfaction & well-being
2. Architectural form
3. Energy savings potential

Benefits of daylight

- Appearance of internal spaces
- Visual interest
- Occupant comfort & satisfaction
- Circadian rhythms
- Energy benefits
- Productivity benefits
- Marketing benefits
- Focus & way-finding
- Biophilia

Sunlight and daylight

✓**Sunlight**
: Direct beam from sun, extremely bright

✓**Daylight**
: Diffuse light from all other parts of the sky, and reflected from other surfaces

Components of daylight

Sunlight

- Direct
- Location and direction predictable as a function of time
- Intensity is a function of sun angles and sky conditions
- Colour temperature is a function of sun angles

Components of daylight

Skylight

- Diffuse
- Location and direction of skylight is influenced by the sun
- Intensity is a function of sun angles and sky conditions
- Colour temperature is a function of sun angles

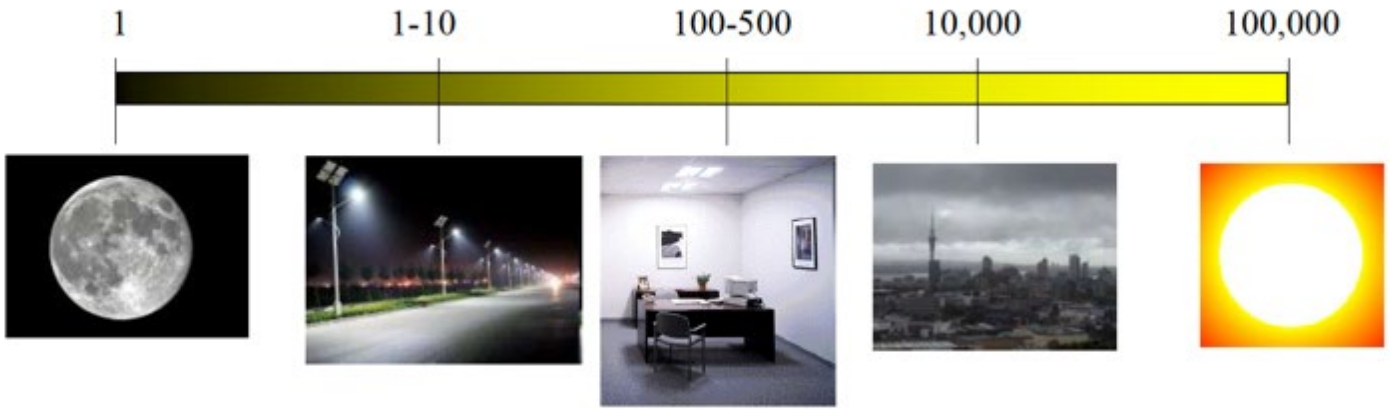
Daylighting Benefits

Daylighting differs from artificial lighting in a number of ways:

- ✓ Spectral content
- ✓ Daylight is diffuse
- ✓ Daylight levels vary
- ✓ Effect on visual comfort and well being
- ✓ Influence energy use
- ✓ **Efficiency** of daylight is considerably higher than electric alternatives: daylight introduces **less heat per lumen** into a building

Sky and sun brightness

Illuminance (Lux)



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Daylighting Benefits

LIGHT SOURCE	EFFICACY*
Direct Sun (low altitude)	90 lm/w
Direct Sun (high altitude)	117 lm/w
Direct Sun (mean altitude)	100 lm/w
Diffuse Sky (clear)	150 lm/w
Diffuse Sky (average)	125 lm/w
Global (average of sky and sun)	115 lm/w
Incandescent (150 w)	16-40 lm/w
Fluorescent (40 w, CWX)	50-80 lm/w
High Pressure Sodium	40-140 lm/w

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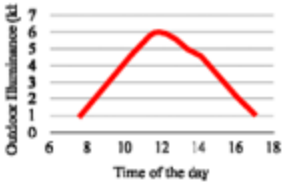
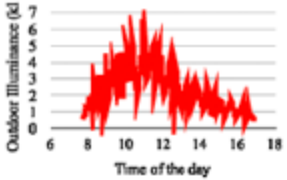
** Efficacy: a measure of how well a light source produces visible light*

Sky Types

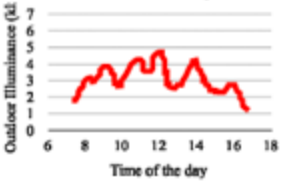
Clear Sky



Partly Cloudy/
Intermediate sky



Overcast



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Windows

- The normal source of interior daylighting
- Significantly affect the level of interior daylight, appearance, visual comfort
- Consider window location, size and construction
- Orientation specific
- User operable shading devices should be available for windows to working spaces

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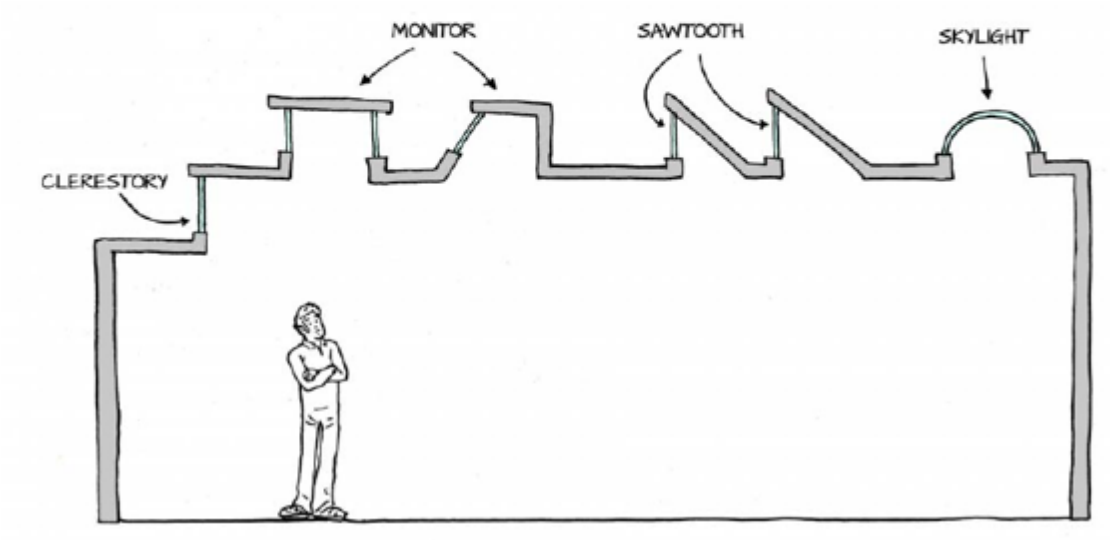
Toplighting



Sainsbury Laboratory, University of Cambridge
Architect: Stanton Williams

Glass types

Glass type	Example application	Rough cost
Single (monolithic)	Residential vision panes	$40-60 / m2
Single (laminated)	Residential low level	$60-90 / m2
Double (with coating)	Commercial	$120-140 / m2
Triple (with coating)	Cold climate	$160-300 / m2
Ceramic frit	Sunlight control	Add $45-70 / m2
Light redirecting	Special applications	$350-550 / m2
Anti-reflective	Display cases	$650-700 / m2
Fire rated	Windows on boundaries	$600-900 / m2

Note that the above rough costs are estimates only, and price will vary strongly based on market, volume & source location

Range of visible transmittances

Generic Glazing Type	Typical Visible Transmittance (%)
Single pane, clear	89
Double pane, clear	80
Double pane, clear, low-e	70
Single pane, blue	57
Double pane, blue	51
Double pane, blue low-e	49
Single pane, grey	42
Double pane, grey	39
Double pane, grey low-e	37
Single pane, reflective	25
Double pane, reflective	20

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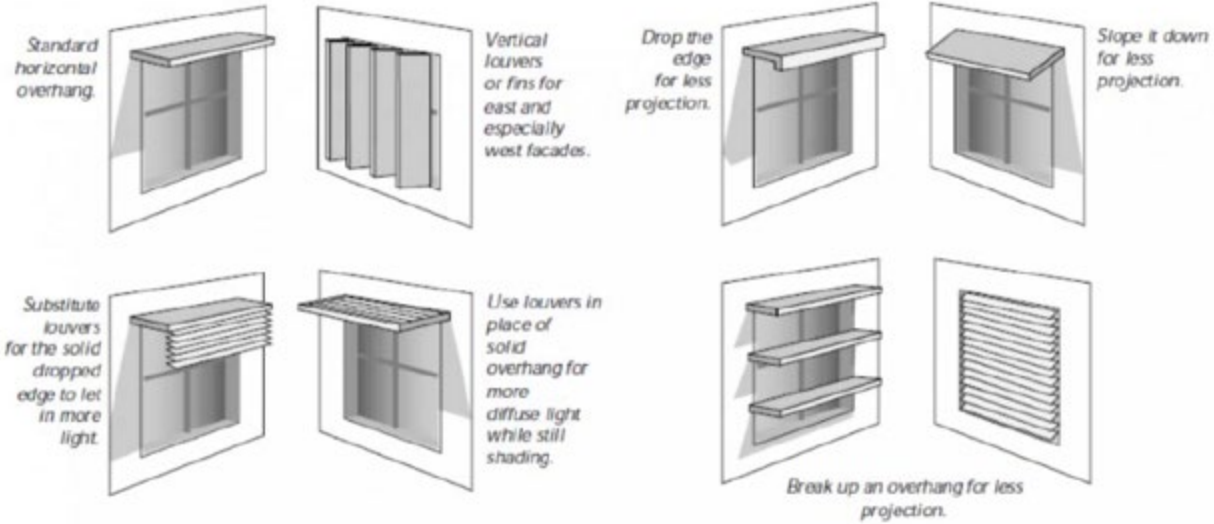
Passive shading



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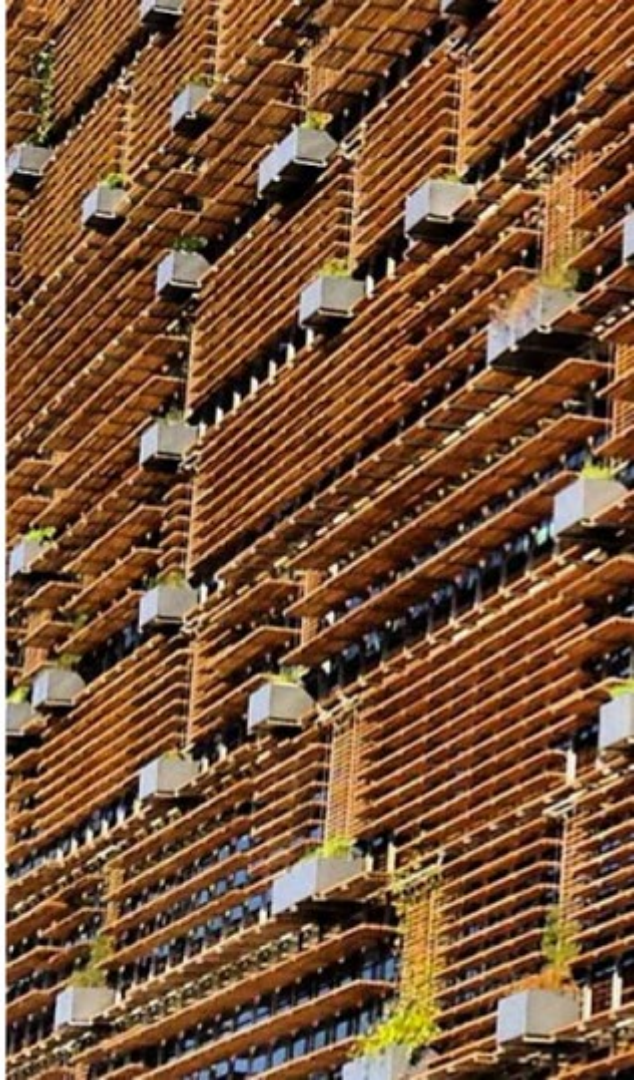
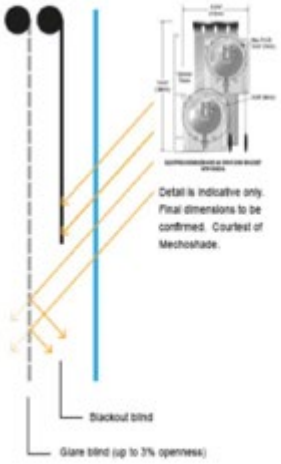
New Acton Nishi, Canberra

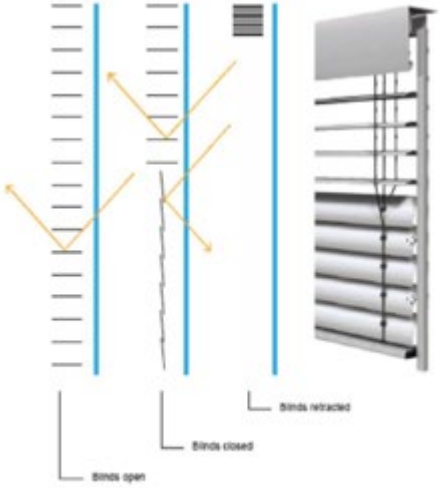


Active shading

Roller shades



Venetian shades



Electrochromic glazing

Los Angeles County Museum of Art (LACMA)

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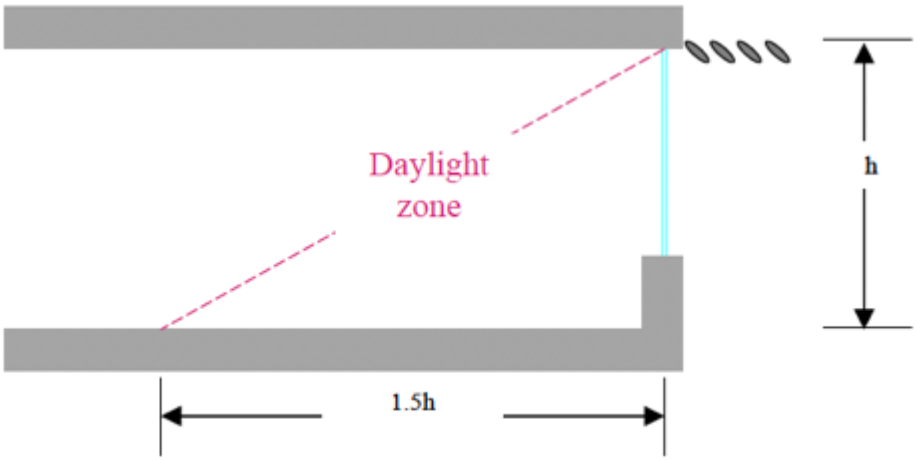
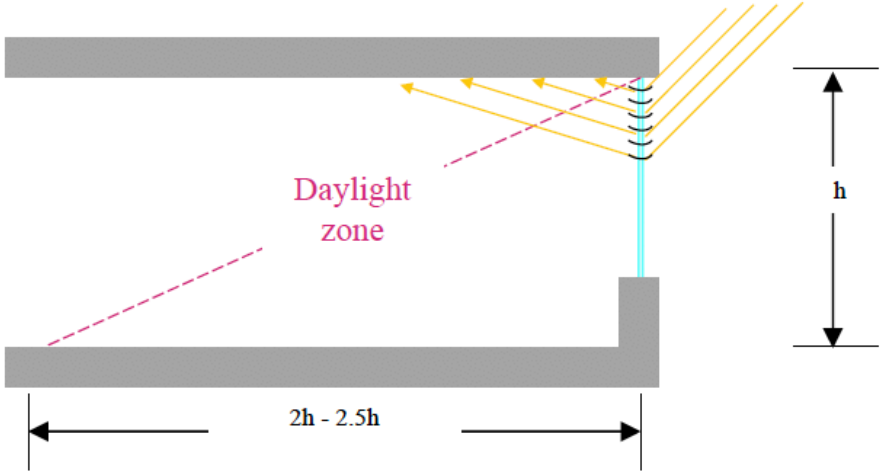
Depth of daylight penetration



Depth of daylight penetration

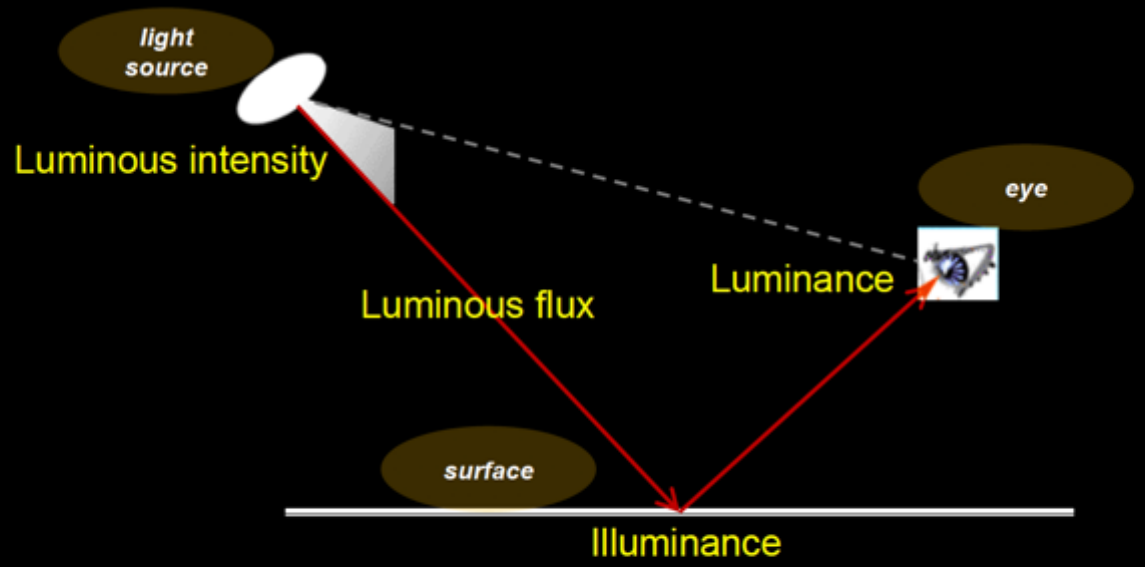


Daylighting Problems

- ✓ Solar gain
- ✓ Visual discomfort: excessive contrast and glare

Photometry

: the science of measurement of visible light in terms of its perceived brightness to human vision



Photometric quantities

- ✓**Luminous Intensity** : Measures photometric light production
  - A measure of the amount of light output by a source
  - Units = *candela* (cd)

- ✓**Luminous Flux** : Measures photometric light transmission
  - The amount of light travelling in some direction.
  - Units = *lumens* (lm)

- ✓**Illuminance** : Measures photometric light incidence
  - The amount of light falling on 1 m^2 of a particular surface.
  - Units = *lux* (lx)

- ✓**Luminance** : Measures photometric light reflection
  - A measure of the brightness of a particular surface
  - Units = candelas per meter squared (cd/m^2)

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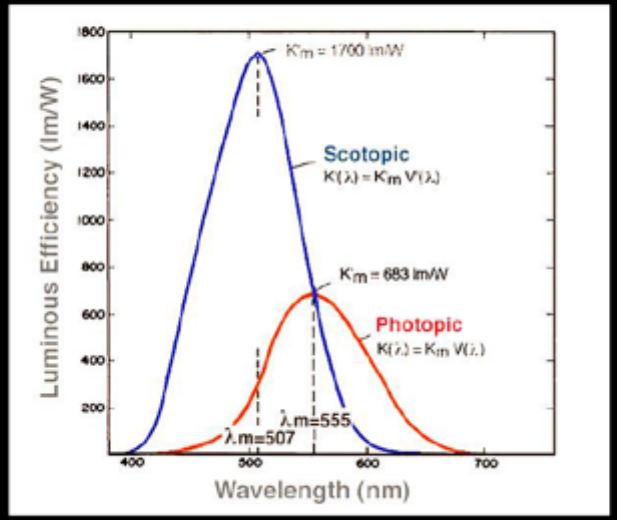
Sensitivity



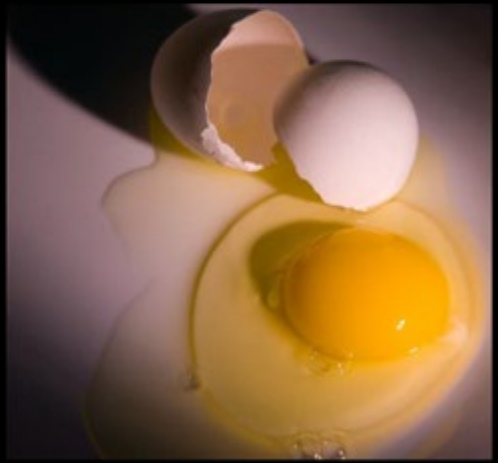
- ✓ **Photopic** vision:
  bright light conditions. Allows colour perception

- ✓ **Scotopic** vision:
  dark adapted state. In this state the eye is far less sensitive to colour

Contrast sensitivity

- ✓ Visual difference in:
  Brightness or
  Colour

- ✓ Directional lighting can reveal form by throwing shadow



Brightness

- ✓ Physical quantity is ‘intensity’ (as measured by a photometer)
- ✓ Subjective response is ‘brightness’ (as perceived, depending on state of adaptation!)

Glare

- ✓Glare occurs whenever one source of light is much brighter than the general brightness in the surroundings
- ✓Common sources of glare:
  - **Luminaires**
  - **Windows**

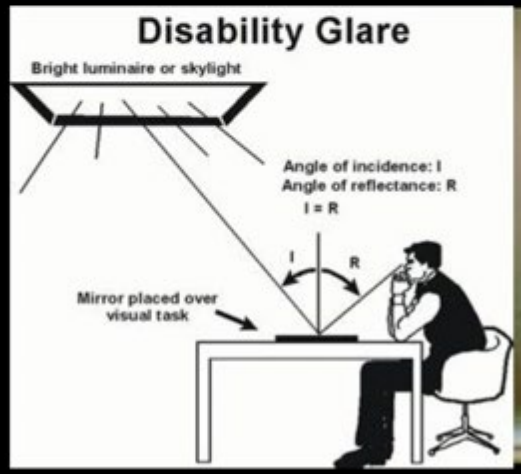
Glare

✓Discomfort or reduction in ability to see significant objects due to:

- Extreme contrast in space or time
- Unsuitable distribution in the range of illuminances in the visual field

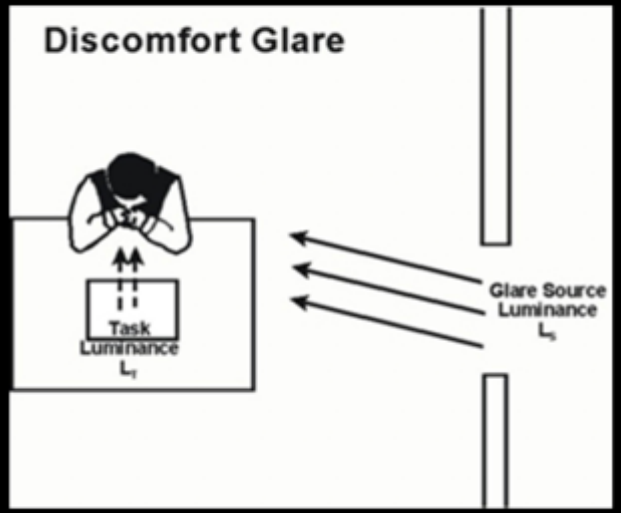
Glare: disability glare

- ✓ Direct reduction in the person's ability to see objects in the field of view
- ✓ e.g. Brilliant light sources, like car headlamps at night, or the view of the sun from a window at the end of a corridor



Glare: discomfort glare

- ✓Glare in which there is no significant reduction in the ability to see, although discomfort still persists, due to the bright sources in the field of view
- ✓e.g. the view of an excessively bright sky near the line of sight of the worker

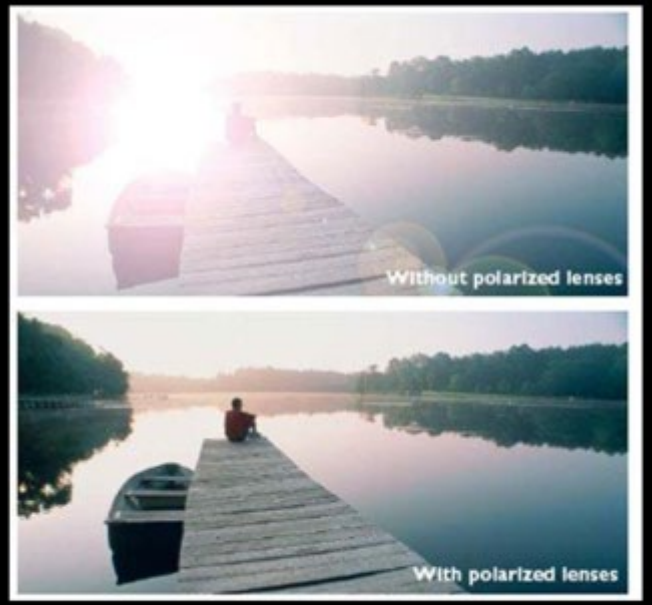


Glare: veiling reflections

- ✓caused when the reflected image of a source of light is brighter than the luminance of the task
- ✓e.g. the image of a window or luminaire off the surface of a computer screen

Glare: reflected glare

- ✓ When light from a light source is reflected off specular surfaces into the eye or field of view
- ✓ e.g. the sun's reflection from a swimming pool



Daylight Factor

A measure of the amount of Daylight indoors

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Daylight factor (DF)

- ✓the ratio of the illuminance at a particular point within an enclosure to the simultaneous unobstructed outdoor illuminance under the same sky conditions
- ✓DF = Indoor illuminance/Outdoor illuminance on an overcast day
- ✓Usually expressed as a percentage (%)

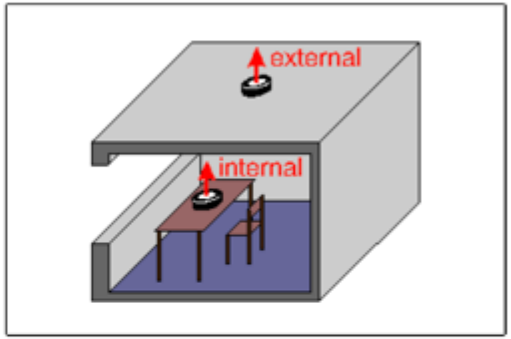

Image source: http://new-learn.info/learn/packages/clear/visual/daylight/analysis/hand/daylight_factor.html

**It has been observed that although overcast sky illuminance may vary between quite wide limits, the ratio between illuminance at a point indoors to that outdoors remains constant (Szokolay, 2008:155)*

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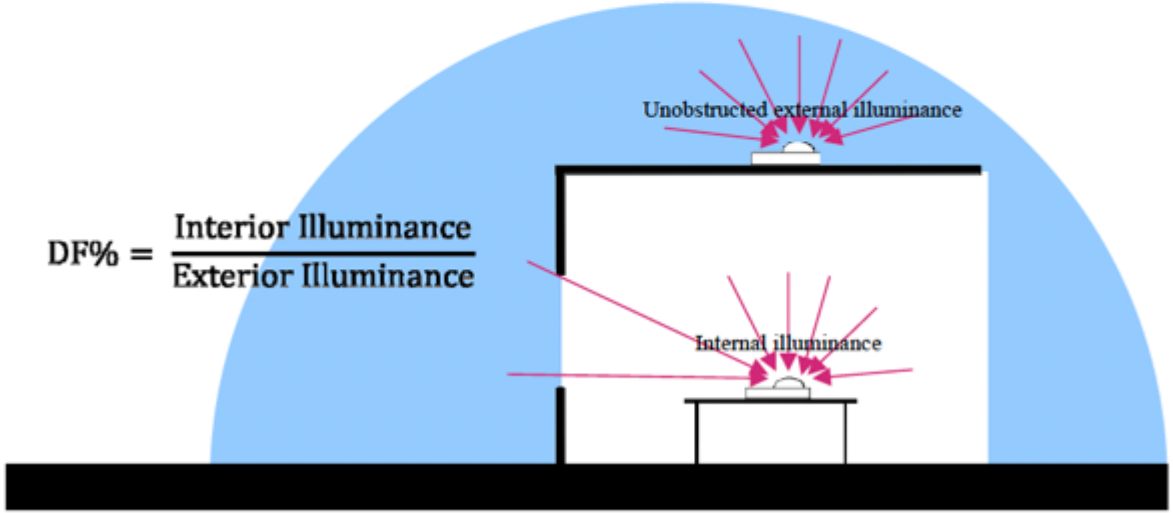
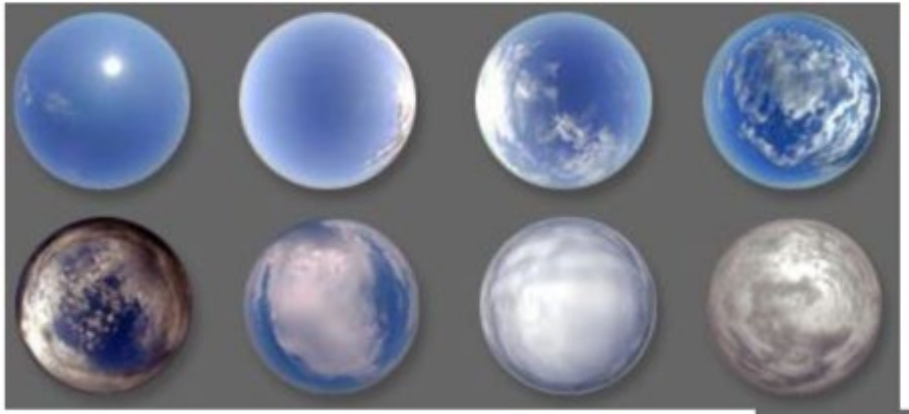
Daylight factor

$$DF\% = \frac{\text{Interior Illuminance}}{\text{Exterior Illuminance}}$$

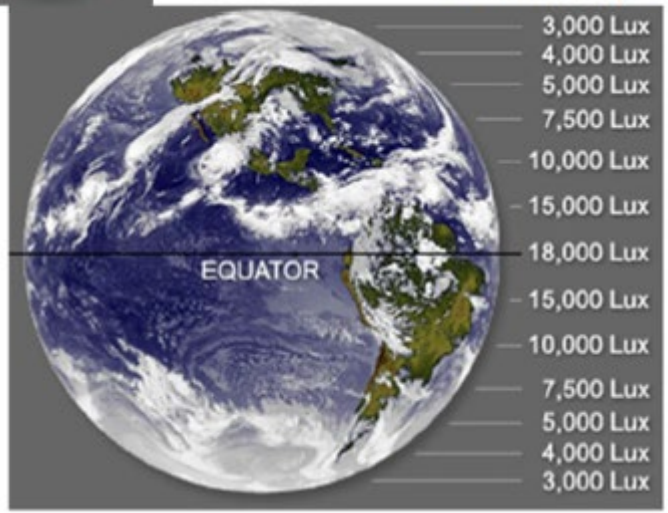


Daylight factor



Sky types

Design sky



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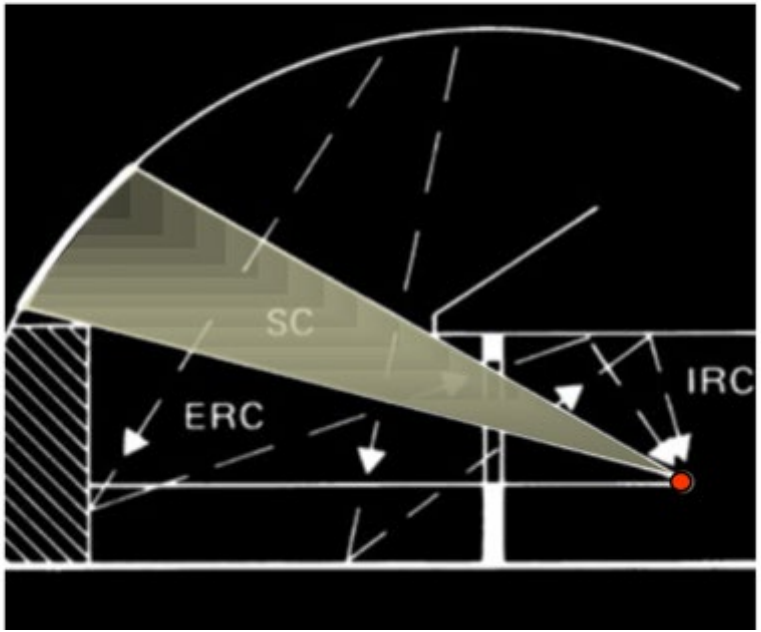
Daylight components



Three main components:

- ✓Sky Component (SC): light from a patch of sky visible from the point considered
- ✓Externally Reflected (ERC): light reflected by outdoor objects
- ✓Internally Reflected (IRC): any light entering the window, not reaching the workplane directly, but only after reflection by internal surfaces

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Daylight Factor

DF	Appearance	Energy implication
< 2%	room looks gloomy	Electric lighting needed most of the day
2 to 5%	Predominantly daylit appearance, but supplementary artificial lighting is needed	Good balance between lighting and thermal aspects
> 5%	Room appears strongly daylit	Daytime electric lighting rarely needed, but potential for thermal problems due to overheating in summer and heat losses in winter

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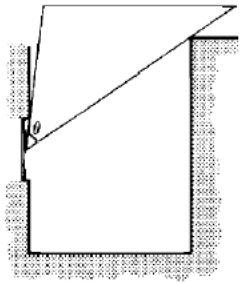
Manual calculation

- W – total glazed area of windows (m2)
- A – total area of all room surfaces: ceiling, floor, walls and windows (m2)
- R – area weighted average reflectance of the room surfaces
- M – maintenance factor
- T – glass light transmission factor (VLT)
- Θ - angle in the vertical plane normal to the window, by sky visible from the centre of the window (between 0 – 90 degrees)

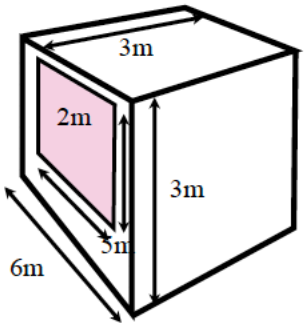
$$\overline{DF} = \frac{M \times W \times \theta \times T}{A\left(1 - R^2\right)} \%$$

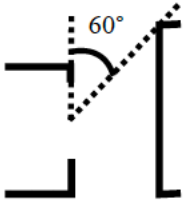


Vertical glazing

Example calculation

- What is the average daylight factor in this simple room?
  - Room dimensions – 6m x 3m x 3m
  - Window dimensions – 5m x 2m
  - Typical reflectance value – 0.5
  - Maintenance Factor – 0.88
  - Glass VLT – 0.70
  - Area of all surfaces – 90 m2
  - Θ - 60 degrees





Example calculation

- What is the average daylight factor in this simple room?
- Average daylight factor formula: $\overline{DF} = \frac{M \times W \times \theta \times T}{A(1 - R^2)}\%$
- M = 0.88
- W = 5m x 2m = 10 m2
- Θ = 60 degrees
- T = 0.7
- A = 2x6x3 + 2x3x3 + 2x6x3 = 90 m2
- R = 0.5

- **Average daylight factor = ??**

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Example calculation

- What is the average daylight factor in this simple room?
- Average daylight factor formula: $\overline{DF} = \frac{M \times W \times \theta \times T}{A(1 - R^2)}\%$
- M = 0.88
- W = 5m x 2m = 10 m2
- Θ = 60 degrees
- T = 0.7
- A = 2x6x3 + 2x3x3 + 2x6x3 = 90 m2
- R = 0.5

- **Average daylight factor = 5.5%**

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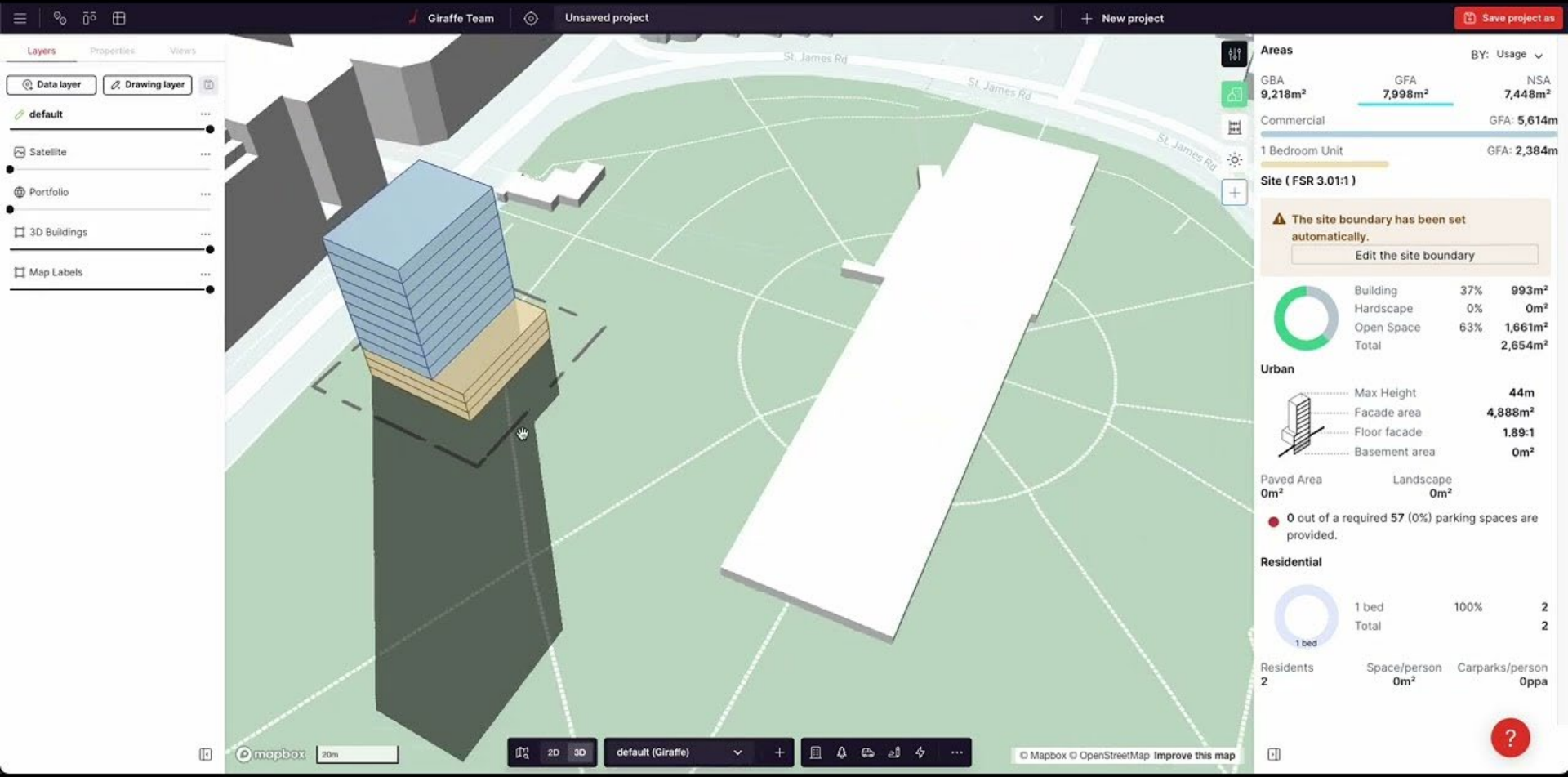
Giraffe Team
Unsaved project
New project
Save project as
Layers
Properties
Views
Data layer
Drawing layer
default
Satellite
Portfolio
3D Buildings
Map Labels
St. James Rd
St. James Rd
St. James Rd
mapbox
20m
2D
3D
default (Giraffe)
© Mapbox © OpenStreetMap Improve this map
Areas
BY: Usage
GBA 9,218m²
GFA 7,998m²
NSA 7,448m²
Commercial GFA: 5,614m
1 Bedroom Unit GFA: 2,384m
Site (FSR 3.01:1)
The site boundary has been set automatically.
Edit the site boundary
Building 37% 993m²
Hardscape 0% 0m²
Open Space 63% 1,661m²
Total 2,654m²
Urban
Max Height 44m
Facade area 4,888m²
Floor facade 1.89:1
Basement area 0m²
Paved Area 0m²
Landscape 0m²
0 out of a required 57 (0%) parking spaces are provided.
Residential
1 bed
1 bed 100% 2
Total 2
Residents 2
Space/person 0m²
Carparks/person 0ppa

Thank you

Next time: Artificial lighting

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